Solution

The equation of the regression line of X on Y is

$$(X-\bar{X}) = r \times \frac{\sigma x}{\sigma y} \times (Y-\bar{Y})$$

Given $\bar{X} = 12$, $\bar{Y} = 10$

$r = 0.85$, $\sigma x = 0.1$ and $\sigma y = 0.2$

Then substituting the values in formula

$$(X-12) = 0.85 \times (0.1/0.2) \times (Y-10)$$
$$(X-12) = 0.85 \times (0.5) \times (Y-10)$$
$$X = 0.425 \times (Y-10) + 12$$
$$X = 0.425Y - 4.25 + 12$$
$$X = 0.425Y + 7.75$$

X on Y

Answer X = 0.425Y + 7.75

The equation of the regression line of Y on X is

$$(Y-\bar{Y}) = r \times \frac{\sigma y}{\sigma x} \times (X-\bar{X})$$

Given $\bar{X} = 12$, $\bar{Y} = 10$

$r = 0.85$, $\sigma x = 0.1$ and $\sigma y = 0.2$

Then substituting the values in formula

$$(Y-10) = 0.85 \times (0.2/0.1) \times (X-12)$$
$$(Y-10) = 0.85 \times (2) \times (X-12)$$
$$Y = 1.7 \times (X-12) + 10$$
$$Y = 1.7X - 20.4 + 10$$
$$Y = 1.7X - 10.4$$

Y on X

Answer Y = 1.7X–10.4

12.14 Introduction To Econometrics

Origin Of Econometrics

Economists tried to support their ideas with facts and figures in ancient times. Irving Fisher is the first person, developed mathematical equation in the quantity theory of money with help of data. Ragnar Frisch, a Norwegian economist and statistician named the integration of three subjects such that mathematics, statistical methods and economics as Econometrics" in 1926.



Ragnar Anton Kittil Frisch Noble Memorial Prize in 1969

The term econometrics is formed from two words of Greek origin, 'oukovouia' meaning economy and 'uetpov' meaning measure. Econometrics emerged as an independent discipline studying economic phenomena.

Econometrics may be considered as the integration of economics, Statistics and Mathematics.

Econometrics is an amalgamation of three subjects which can be easily understood by following Venn diagram and picture representation.

Economics + Mathematics= Mathematical Economics

Mathematical Economics+ Statistical Data & Its Technique = Econometrics

{Economics + Statistics + Mathematics}+Empirical Data = Econometrics

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